Exploring PatchTST: A Transformer Approach to Time Series Forecasting

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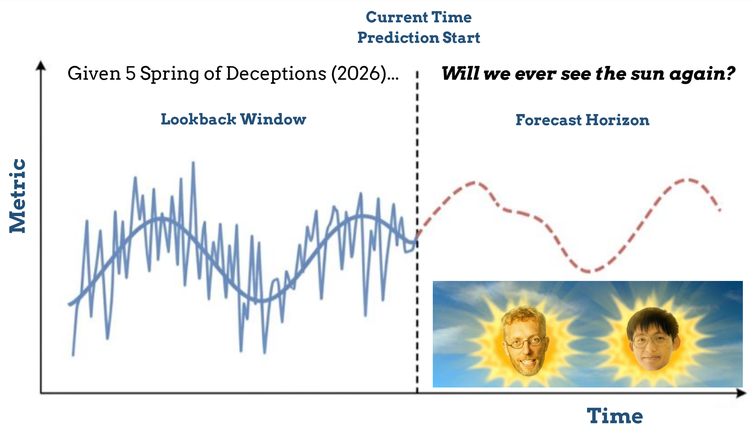
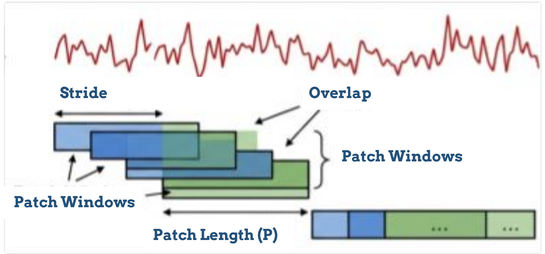
Introduction



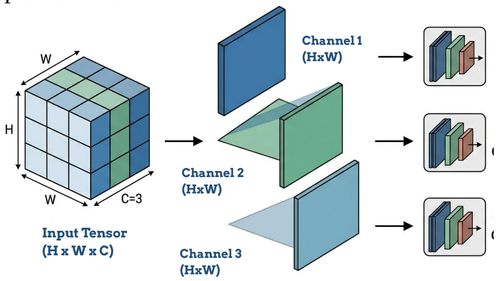
Background and Motivation

- Transformer models applied to time series tasks were being beat by linear models [2] They are different from traditional NLP tasks
- Our goal is to show that with patching and channel independence, we can show that transformer models outperform linear models on time series tasks

1. Patching

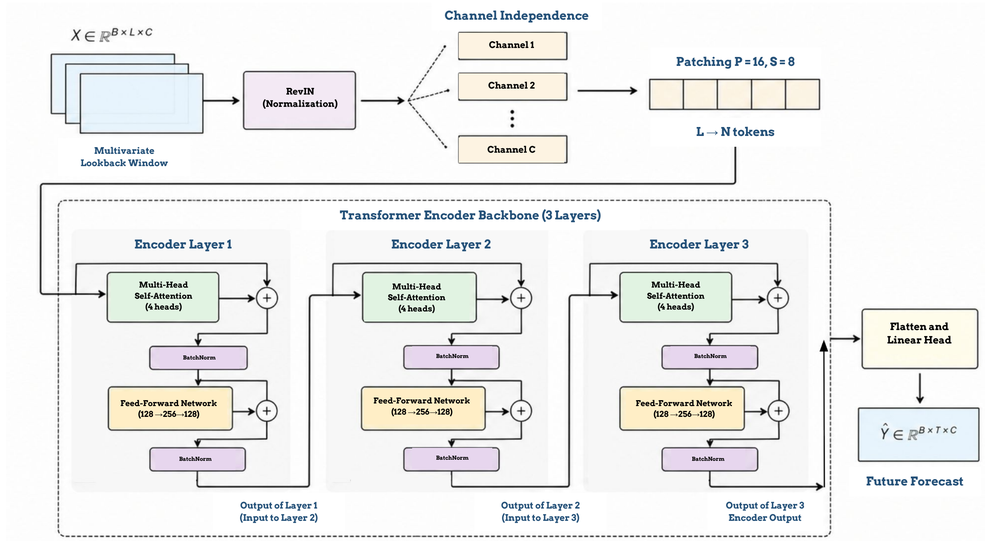


2. Channel Independence

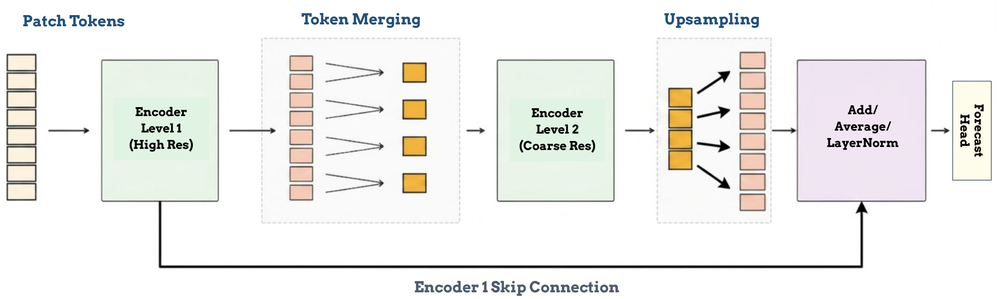


Methods

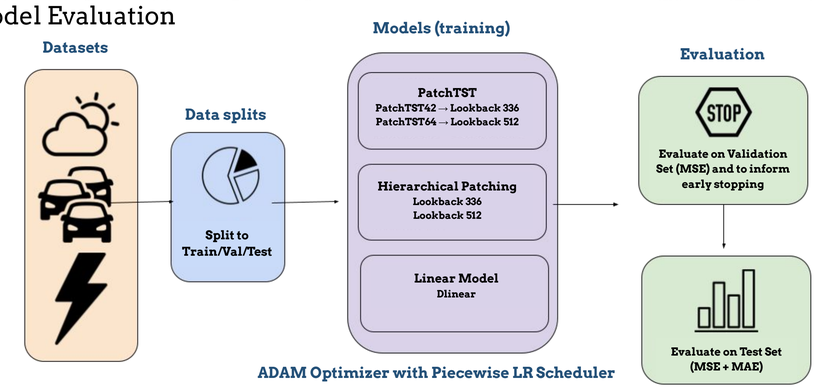
Architecture



Hierarchical Patching Independent Exploration (HPatchTST)



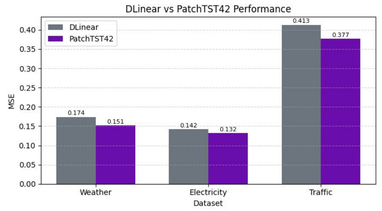
Model Evaluation



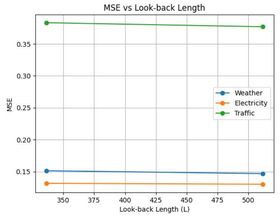
Results

	Model	Dataset					
	Pred Len = 96	Weather		Electricity		Traffic	
		MSE	MAE	MSE	MAE	MSE	MAE
Ours	**dLinear**	0.1740	0.2349	0.1416	0.2398	0.4129	0.2888
	PatchTST/42 (L=336)	0.1513	0.1997	0.1317	0.2258	0.3830	0.2618
	PatchTST/64 (L=512)	**0.1468**	**0.1963**	0.1302	0.2249	**0.3767**	0.2602
	HPatch/42 (L=336)	0.1495	0.1970	0.1319	0.2254	0.3822	0.2643
	HPatch/64 (L=512)	0.1478	0.1988	0.1301	0.2238	**0.3767**	0.2627
Paper	**PatchTST/42 (L=336)**	0.1520	0.1990	0.1300	**0.2220**	0.3670	0.2510
	PatchTST/64 (L=512)	0.1490	0.1980	**0.1290**	**0.2220**	0.3670	**0.2490**

PatchTST outperforms Linear Models



Models improve with larger look back (L)

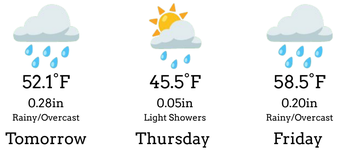


Conclusion

Takeaways:

- PatchTST (and HPatchTST) perform better than linear models
- Reduction in time and space complexity (N = L/S)
- Hierarchical model gains are insignificant from base PatchTST
  - Marginal gains in specific datasets, warrants further study

So will we see the sun?



No, we won't :(

Future work:

- Modelling cross channel dependencies
- Adaptive patching

References:

[1] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2022. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. *arXiv:2211.14730*.

[2] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? *arXiv preprint arXiv:2205.13504*, 2022.